EC519 Project Emotion Recognition in Speech Che Yu

Our goal is to recognise the emotion expressed in a short speech recording. We compare three classical signal-processing feature representations for speech, MFCC, LPC, and LPCC, all fed into the same hidden Markov model (HMM) classifier so that the comparison isolates the effect of the feature choice rather than the model. We used RAVDESS speech dataset sampled at $f_s = 16\,\text{kHz}$, classified into five emotions *neutral, happy, sad, angry, fearful*, with a speaker-independent train/test split so the model never sees a test speaker during training.

General Pipeline

Every feature set follows the same flow: the raw waveform is cut and windowed into short overlapping frames, each frame is converted into a feature vector. The frame features are then turned into a sequence of discrete symbols, and an HMM is trained to predict the five emotion classes and model those sequences. Our pipeline is shown below:

$$\text{audio} \longrightarrow \text{framing + windowing} \longrightarrow \text{feature extraction} \longrightarrow \text{VQ symbols} \longrightarrow \text{HMM per class.}$$

(a) Preprocessing: framing and windowing. Speech is non-stationary on the scale of phonemes (50–200 ms), but *quasi-stationary* inside short windows of about 25 ms. Since we are not modeling phonetic detail, we can use this stationarity assumption to slice the waveform into frames of $L_w = 25\,\text{ms}$ with a hop of $L_h = 10\,\text{ms}$ (so consecutive frames overlap by 60%). At 16 kHz this gives frames of $N = 400$ samples spaced by $H = 160$ samples. Then each frame is multiplied by a Hamming window $w[n] = 0.54 - 0.46\cos\big(2\pi n/(N-1)\big)$, which tapers the edges so the spectrum is not contaminated by the abrupt discontinuities at the frame boundaries.

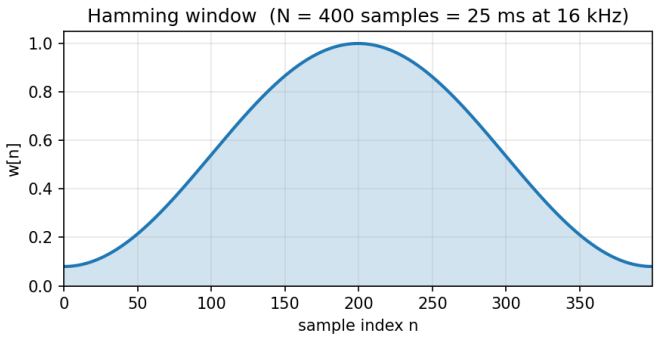

Figure 1: *
Hamming window ($N = 400$ samples $= 25\,\text{ms}$ at $16\,\text{kHz}$).

(b) Feature extraction. Each extractor turns a windowed frame into a fixed-length feature vector. The three feature families capture the speech signal in different ways:

MFCC (Mel-Frequency Cepstral Coefficients). A perceptually motivated representation of the short-time spectral envelope. The frame is sent through the following pipeline:

$$s_t^w[n] \longrightarrow |\text{FFT}|^2 \longrightarrow \text{mel filterbank} \longrightarrow \text{log} \longrightarrow \text{DCT-II} \longrightarrow \mathbf{c}_t.$$

The power spectrum is summed through a bank of triangular filters spaced on the mel scale (denser at low frequencies, where speech information lives), the log of each band energy is taken, and an orthonormal DCT-II compresses the 26 log-band energies down to the first 12 cepstral coefficients which describe the smooth shape of the vocal-tract spectrum (as shown in Lecture 19). We additionally append the log frame energy and stack first- and second-order time differences (Δ and $\Delta\Delta$) so each frame also encodes how the spectrum is changing locally, giving a 39-D vector.

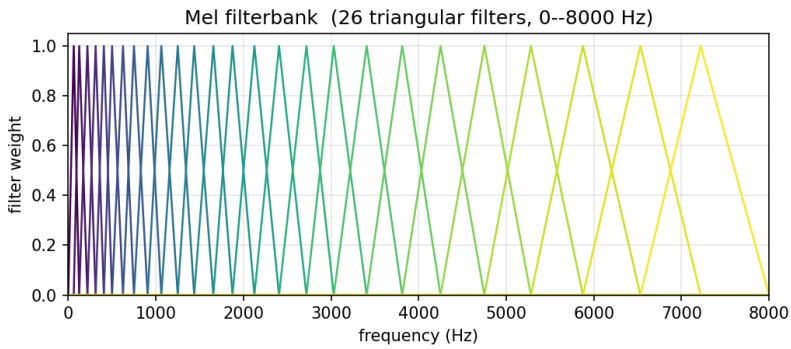

Figure 2: *
The 26-filter triangular mel filterbank applied to the power spectrum.

LPC (Linear Predictive Coding). Each frame is modelled as the output of an all-pole filter that captures the resonances of the vocal tract:

$$H(z) = \frac{G}{A(z)}, \qquad A(z) = 1 + \sum_{k=1}^{p} a_k\, z^{-k}.$$

The coefficients $a_1, \ldots, a_p$ are obtained from the frame's autocorrelation via the Levinson–Durbin recursion. We use $p = 12$ and append the log of the prediction-error energy as a loudness/voicing cue, giving a 13-D vector.

LPCC (LP Cepstral Coefficients). The cepstrum derived analytically from the LPC coefficients we've obtained, with $\alpha_k = -a_k$:

$$c_m = \alpha_m + \sum_{k=1}^{m-1} \frac{k}{m}\, c_k\, \alpha_{m-k}, \qquad 1 \le m \le p.$$

LPCCs retain qualities of the all-pole model with the better numerical and statistical properties of cepstral features. It also deconvolves the vocal-tract response from the excitation, so it may be more robust to variations in pitch and preserve perceptual characteristics of speech (as in Lecture 16&17). We keep $c_1, \ldots, c_{12}$ and append first-order time differences, giving a 24-D vector.

(c) HMM classification. For each feature set we first fit a vector quantiser (k-means with $V = 16$ centroids) on the stacked training frames and use it to encode every utterance into a sequence of discrete symbols $O = (o_1, \ldots, o_T)$, $o_t \in \{1, \ldots, V\}$. We then train one discrete HMM per emotion class c, $\lambda_c = (\pi^c, A^c, B^c)$, with $K = 3$ hidden states using Baum–Welch (EM) algorithm (as in Lecture 19&20). At test time the predicted emotion is the class whose model assigns the highest likelihood to the symbol sequence: $\hat{y} = \arg\max_c \log P(O \mid \lambda_c)$.

Results

Test accuracies of our different feature sets are shown below (five classes, chance = 20%):

Feature set	Frame dim D	Test accuracy
LPC	13	**0.367**
MFCC	39	0.356
LPCC	24	0.278
Combined	65	0.250

LPC and MFCC are essentially tied at the top; LPCC and the combined set trail. The combined set's relative weakness is consistent with the mismatch between its 65-D feature space and a 16-symbol codebook: too few symbols to preserve class-discriminative structure in a much larger space.

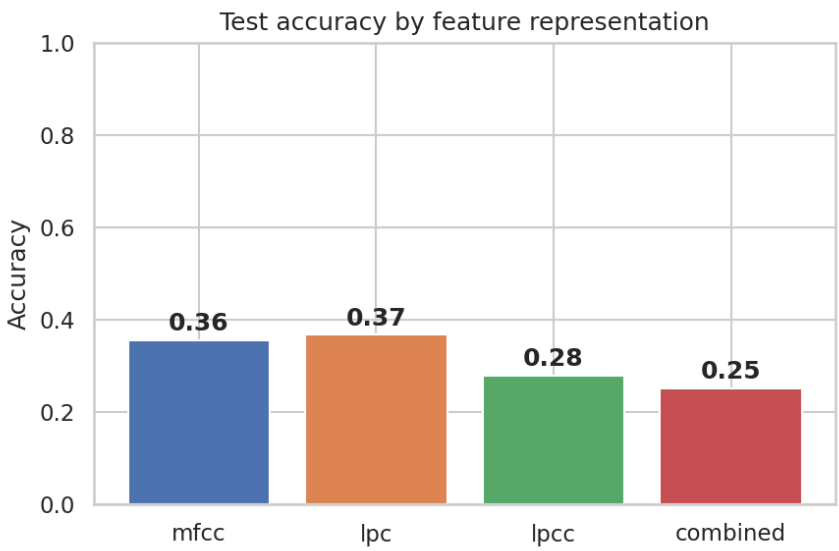

Figure 3: *
Test accuracy by feature set.

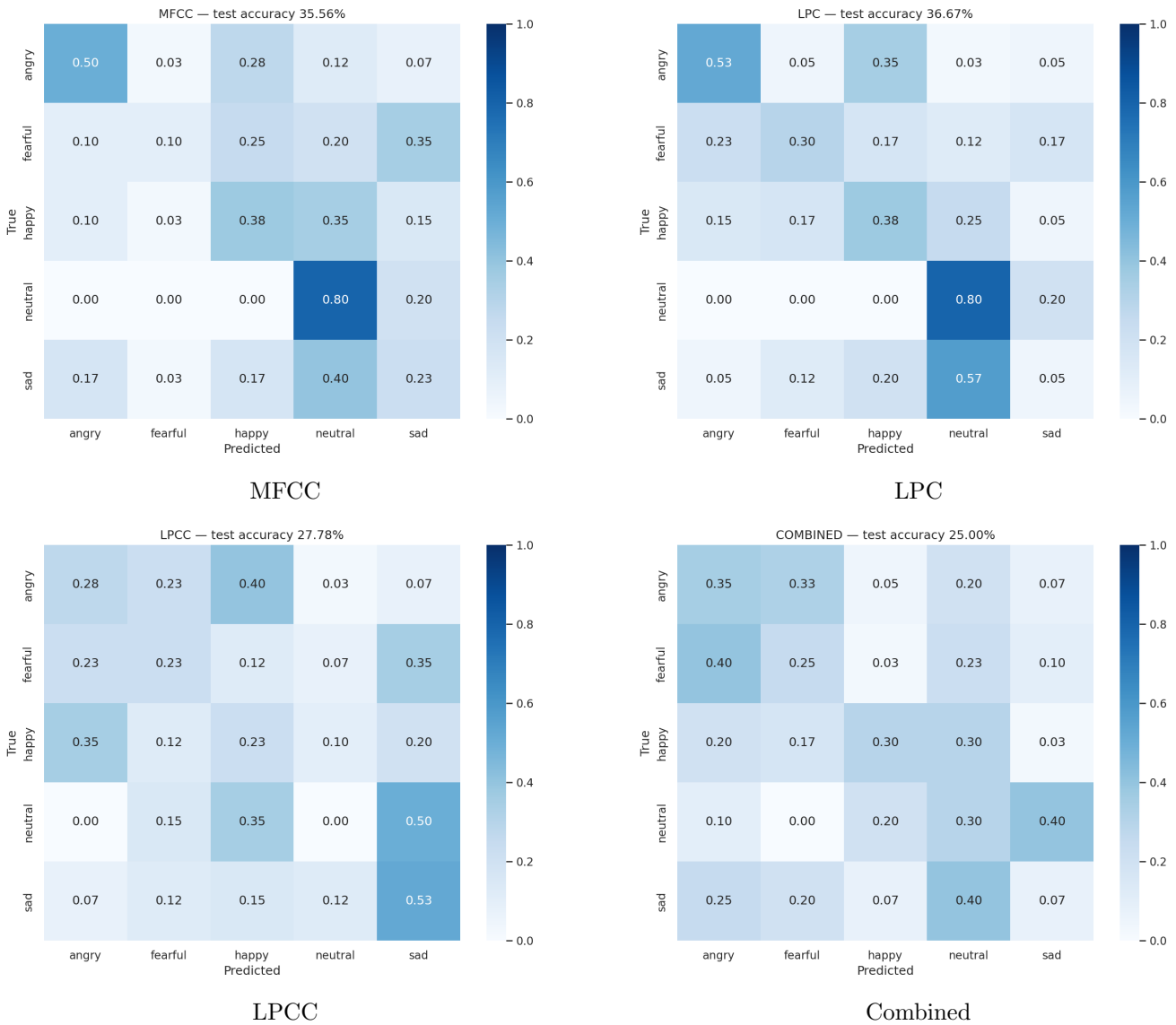

Figure 4: *
Per-feature confusion matrices (normalised by true class).

Future Work

This project utilizes the whole EC519 lecture pipeline and I think it is an interesting project since LLM (Whisper) have not gain the ability to understand the emotion of the speech. However, the results are not very good, and there is a lot of room for improvement. Several directions follow naturally from the results above. One of the problems that the combined feature set under-performs because a $V = 16$ codebook is too coarse for a 65-D space; sweeping $V \in \{32, 64, 128\}$ might address this. Another aspect is that, throughout this project, I am intrigued by the potential of output vectorzied classification rather than probability like HMM does. Vectorized classification can be modeled into 6 dimensional space for basic emotion (happy, sad, angry, fearful, suprise, disgust). And that we can get rid of neutral, calm, and other non-basic emotion, which are more difficult to classify. Third, our feature menu is purely spectral / source-filter; adding *prosodic* cues (pitch contour, energy envelope, speaking rate) is well-motivated for emotion (e.g. anger and joy expand pitch range while sadness compresses it). Finally, RAVDESS is a small, acted, studio-clean dataset, so the

absolute accuracies are likely optimistic. Evaluating on a more naturalistic dataset such as CREMA-D would test how much of the signal is genuine emotion.

References

References

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Code

The full implementation is open-source and available at:

`https://github.com/cheyu-cyber/Emotional_Recognition_in_Speech`